Assignment – 2.14

Build and Configure a Small Office Network

Q - 1. Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

- **Assign Static IP Addresses to Five Devices:**
- **Steps:**

1. Connect five computers to the same network.
2. Open the network settings of each computer.
3. Select IPv4 and choose Manual IP configuration.
4. Assign the following IP addresses.

Device	IP Address	Subnet Mask
PC 1	192.168.1.10	255.255.255.0
PC 2	192.168.1.11	255.255.255.0
PC 3	192.168.1.12	255.255.255.0

Device	IP Address	Subnet Mask
PC 4	192.168.1.13	255.255.255.0
PC 5	192.168.1.14	255.255.255.0

➢ Verification:

- Open Command Prompt.
- Type ping 192.168.1.11
- Check whether you receive replies.

➢ Result:

All five devices have static IP addresses and can communicate with each other.

Q - 2. Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

- **Set Up a DHCP Server:**
- **Steps:**

1. Open Cisco Packet Tracer.
2. Add one router or DHCP server and three PCs.
3. Configure the DHCP server with:
    - Start IP: 192.168.1.100
    - Subnet Mask: 255.255.255.0
    - Default Gateway: 192.168.1.1
4. Enable the DHCP service.
5. Select DHCP in each PC's IP configuration.
6. Check the IP address assigned to each PC.

Device	Assigned IP
PC 1	192.168.1.100

Device	Assigned IP
PC 2	192.168.1.101
PC 3	192.168.1.102

- **Result:**

  All three devices automatically receive unique IP addresses from the DHCP server.

Q - 3. Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface. Hint: You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.

- **Configure DNS Service**
- **Steps:**

1. Open Cisco Packet Tracer.
2. Add a server and configure its IP address as 192.168.1.10.
3. Open the DNS service.
4. Turn DNS ON.
5. Add a DNS record:
    - Name: musicplayer.local
    - Address: 192.168.1.10
6. Enable the HTTP service on the server.
7. Open a PC's web browser.
8. Type http://musicplayer.local.

- **Result:**

The local web page opens using the custom hostname.

Q - 4. Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN. Hint: Use ping or file sharing to test connectivity between devices in different VLANs.

- **Create Two VLANs:**
- **Steps:**

1. Open Cisco Packet Tracer.
2. Add one switch and four PCs.
3. Create two VLANs on the switch.

VLAN	Name	Devices
VLAN 10	Workstations	PC1, PC2
VLAN 20	Smart Devices	PC3, PC4

4. Assign PC1 and PC2 switch ports to VLAN 10.
5. Assign PC3 and PC4 switch ports to VLAN 20.
6. Test connectivity using the ping command.

➢ Testing:

- PC1 to PC2: Successful.
- PC3 to PC4: Successful.
- PC1 to PC3: Failed.

➢ Result:

Devices in the same VLAN can communicate, but devices in different VLANs cannot communicate without inter-VLAN routing.

Q - 5. Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

- **Configure NAT (Network Address Translation):**
- **Steps:**

1. Open Cisco Packet Tracer.
2. Add a router, switch and two PCs.
3. Connect the devices.
4. Assign private IP addresses to the PCs.
5. Configure NAT on the router.
6. Configure the router's external interface with a public or simulated external IP address.
7. Enable NAT overload (PAT).
8. Test internet connectivity.

- **Testing:**

Open Command Prompt and type:

ping 8.8.8.8

You can also test:

ping www.wikipedia.org